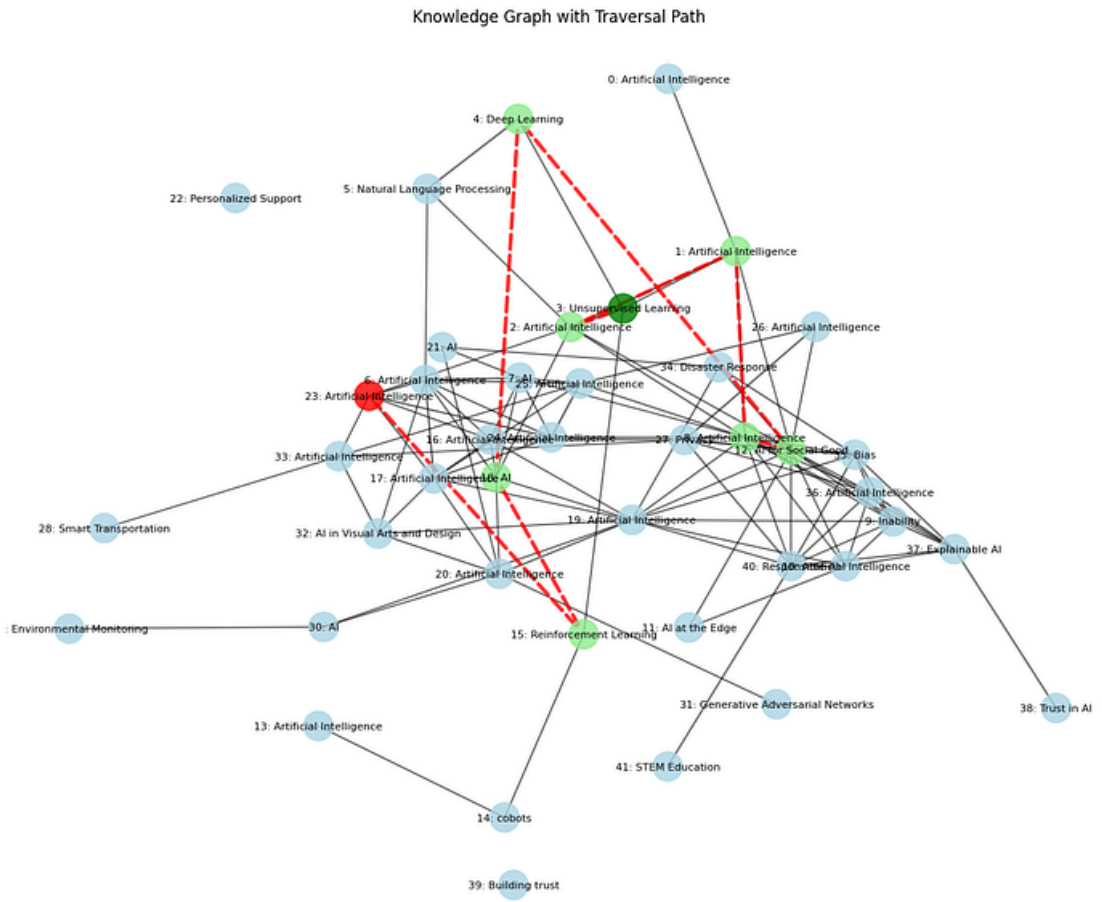


Testing 18 RAG Techniques to Find the Best

crag, HyDE, fusion and more!

Level Up Coding a11y-light ~36 min read · March 12, 2025 (Updated: March 12, 2025) · Free: No

Read this story for free: link

We are going to start with a simple RAG approach, which we all know, and then test more advanced techniques like CRAG, Fusion, HyDE, and more!

To keep everything simple...

> I didn't use LangChain or FAISS

But use only basic libraries to code all the techniques in **Jupyter notebook style** to keep things simple and learnable.

GitHub Repo:

All the step-by-step notebooks are available here:

Implementation of all RAG techniques in a simpler way - FareedKhan-dev/all-rag-techniques

github.com

Codebase is organized as follows:

Copy

```
├── 1_simple_rag.ipynb
├── 2_semantic_chunking.ipynb
...
├── 9_rse.ipynb
├── 10_contextual_compression.ipynb
├── 11_feedback_loop_rag.ipynb
├── 12_adaptive_rag.ipynb
...
├── 17_graph_rag.ipynb
├── 18_hierarchy_rag.ipynb
├── 19_HyDE_rag.ipynb
├── 20_crag.ipynb
└── data/
    └── val.json
    └── AI_information.pdf
    └── attention_is_all_you_need.pdf
```

Table of contents

1. Test Query and LLMs
2. Technique that works best!
3. Importing Libraries
4. Simple RAG
5. Semantic Chunking
6. Context Enriched Retrieval
7. Contextual Chunk Headers
8. Document Augmentation
9. Query Transformation
10. Re-Ranker
11. RSE
12. Contextual Compression
13. Feedback Loop
14. Adaptive RAG
15. Self RAG

17. Hierarchical Indices
18. HyDE
19. Fusion
20. Multi Model
21. Crag
22. Conclusion

Test Query and LLMs

To test each technique, we need four things:

1. Test query and its true answer.
2. PDF document on which RAG will be applied.
3. Embedding generation model.
4. Response and validation LLM.

Using the **Claude 3.5 Thinking model**, I have created a **16+ page long document** on AI topic as a reference document for RAG and **Attention is all you need** paper

For **response generation and validation**, we will use **LLaMA-3.2–3B Instruct** to test how well a tiny LLM can perform for the RAG task.

For **embeddings**, we will be using the **TaylorAI/gte-tiny** model.

Our test query is a **complex one** that we will use throughout the document, and its true answer is:

```
Copy

test query:
How does AI's reliance on massive data sets act as a double-edged sword?

True Answer:
It drives rapid learning and innovation while also
risking the amplification of inherent biases,
making it crucial to balance data volume with fairness and quality.
```

(Conclusion) Technique that works best!

Instead of providing it at the end, it's better to write it at the top, After testing 18 different RAG techniques across our test query.

By intelligently classifying query and selecting the most appropriate retrieval strategy for each question type, Adaptive RAG shows better performance over other approaches. The ability to dynamically switch between factual, analytical, opinion, and contextual strategies allows it to handle diverse information needs with remarkable accuracy.

While techniques like Hierarchical Indices (0.84), Fusion (0.83), and CRAG (0.824) also performed admirably, Adaptive RAG's flexibility gives it the edge in real-world applications.

Importing Libraries

Let's clone my repo first so to install the required dependencies and start working.

Copy

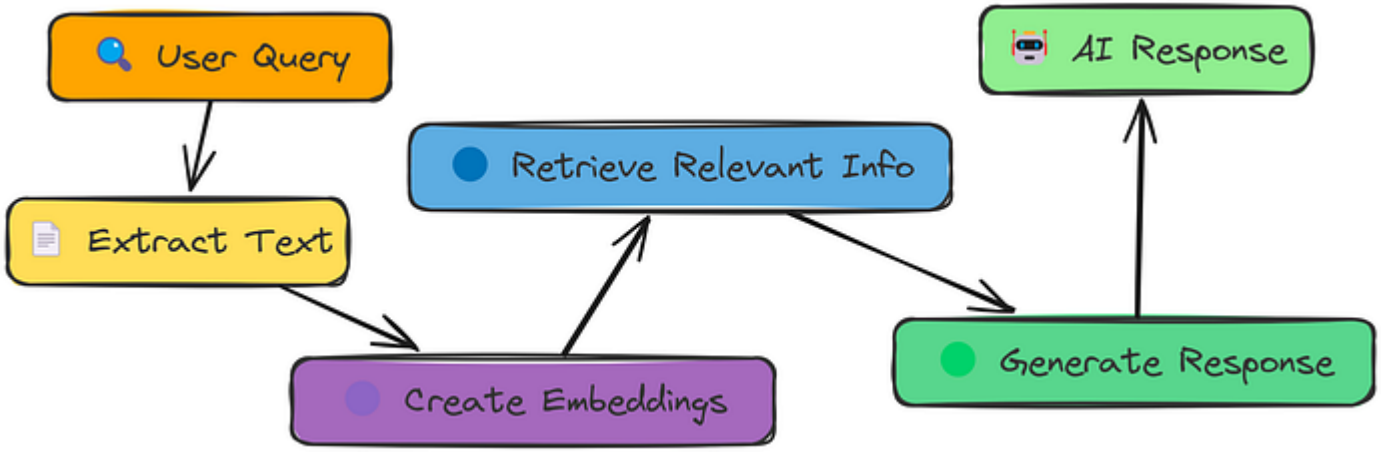
```
# Cloning the repo
git clone https://github.com/FareedKhan-dev/all-rag-techniques.git
cd all-rag-techniques
```

Installing the required dependencies.

```
# Installing the required libraries
pip install -r requirements.txt
```

Simple RAG

Let's start with the simplest RAG. First, we will visualize how it works, and then we will test and evaluate it.



Simple RAG workflow (Created by Fareed Khan)

As the diagram shows, a Simple RAG pipeline works as follows:

- Extract text from a PDF.

- Convert the chunks into numerical embeddings.
- Search for the most relevant chunks based on a query.
- Generate a response using the retrieved chunks.
- Compare the response with the correct answer to evaluate accuracy.

First, let's load our document, grab the text, and split it into manageable chunks:

Copy

```
# Define the path to the PDF file
pdf_path = "data/AI_information.pdf"

# Extract text from the PDF file, and create smaller, overlapping chunks.
extracted_text = extract_text_from_pdf(pdf_path)
text_chunks = chunk_text(extracted_text, 1000, 200)

print("Number of text chunks:", len(text_chunks))


### OUTPUT ###
Number of text chunks: 42
```

This code uses `extract_text_from_pdf` to pull all the text out of our `PDF file`. Then, `chunk_text` breaks that big block of text into smaller, overlapping pieces,

Next, we need to turn those `text chunks` into `numerical representations (embeddings)`:

```
Copy

# Create embeddings for the text chunks
response = create_embeddings(text_chunks)
```

Here, `create_embeddings` takes our list of `text chunks` and uses our embedding model to generate a `numerical embedding` for each one. These embeddings capture the `meaning of the text`.

Now we can perform a `semantic search`, finding the `chunks most relevant` to our test query:

```
Copy

# Our test query, and perform semantic search.
query = '''How does AI's reliance on massive data sets act
           as a double-edged sword?'''
top_chunks = semantic_search(query, text_chunks, embeddings, k=2)
```

With our `relevant chunks` in hand, let's generate a `response`:

```
Copy

# Define the system prompt for the AI assistant
system_prompt = "You are an AI assistant that strictly answers based on the giv

# Create the user prompt based on the top chunks, and generate AI response.
user_prompt = "\n".join([f"Context {i + 1}:\n{chunk}\n========\n" for i, chunk
user_prompt = f"{user_prompt}\nQuestion: {query}"
ai_response = generate_response(system_prompt, user_prompt)
print(ai_response.choices[0].message.content)
```

This code formats the `retrieved chunks` into a `prompt` for a `large language model (LLM)`. The `generate_response` function sends this prompt to the `LLM`, which crafts an `answer` based only on the provided `context`.

Finally, let's see how well our `simple RAG` performed:

```
Copy

# Define the system prompt for the evaluation system
evaluate_system_prompt = "You are an intelligent evaluation system tasked with
```

```
evaluation_prompt = f"User Query: {query}\nAI Response:\n{ai_response.choices[0
evaluation_response = generate_response(evaluate_system_prompt, evaluation_pro
print(evaluation_response.choices[0].message.content)



### OUTPUT ###
... Therefore, the score of 0.3 being not very close to the
true response, and not perfectly aligned.
```

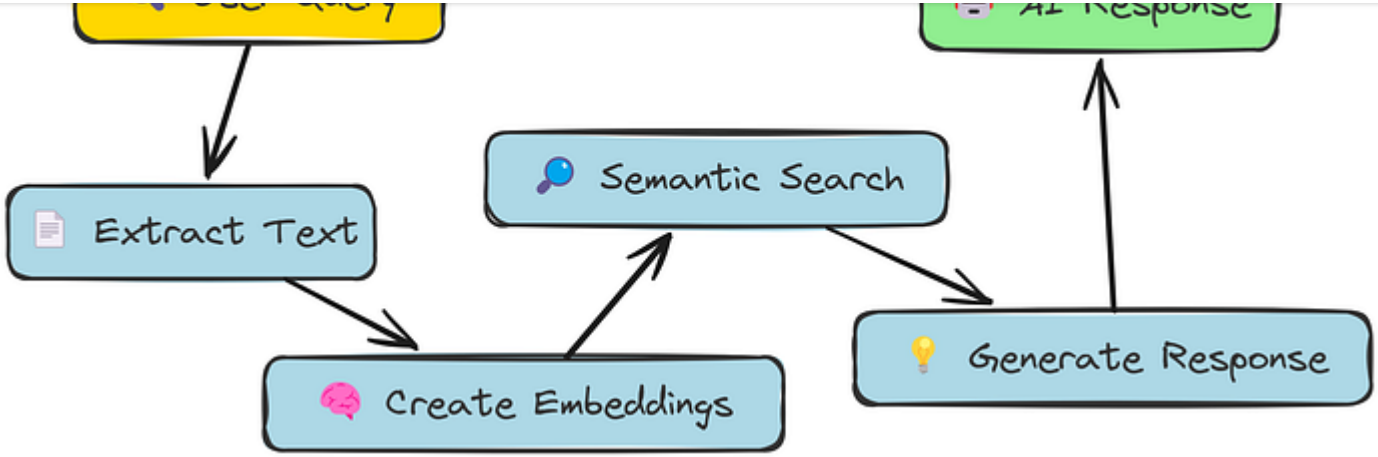
> hmm... the response of simple rag is below average

Let's move on to our next approach.

Semantic Chunking

In our Simple RAG approach, we just chopped the text into fixed-size chunks. This is pretty crude! It might split a sentence in half, or group unrelated sentences together.

Semantic Chunking aims to be smarter. Instead of fixed sizes, it tries to split the text based on **meaning**, grouping semantically related sentences together.



Semantic Chunking Workflow (Created by Fareed Khan)

The idea is, if sentences are talking about similar things, they should be in the same chunk. We'll use the same embedding model to figure out how similar sentences are.

Copy

```
# Splitting text into sentences (basic split)
sentences = extracted_text.split(". ")

# Generate embeddings for each sentence
embeddings = [get_embedding(sentence) for sentence in sentences]

print(f"Generated {len(embeddings)} sentence embeddings.")
```

```
255
```

This code splits our extracted_text into individual sentences. Then create embeddings for each individual sentence.

Now, we'll calculate the similarity between *consecutive* sentences:

Copy

```
# Compute similarity between consecutive sentences
similarities = [cosine_similarity(embeddings[i], embeddings[i + 1]) for i in ra
```

This `cosine_similarity` function (which we defined earlier) tells us how similar two `embeddings` are. A score of `1` means they're `very similar`, and `0` means they're `completely different`. We calculate this score for each pair of `adjacent sentences`.

`Semantic chunking` is deciding where to `split the text into chunks`. We'll use a `"breakpoint"` method. We use the `percentile method` here, looking for `big drops in similarity`:

Copy

The `compute_breakpoints` function, using the `"percentile"` method, identifies points where the `similarity` between sentences `drops significantly`. These are our `chunk boundaries`.

Now we can create our `semantic chunks`:

```
Copy

# Create chunks using the split_into_chunks function
text_chunks = split_into_chunks(sentences, breakpoints)
print(f"Number of semantic chunks: {len(text_chunks)}")


### OUTPUT ###
Number of semantic chunks: 145
```

`split_into_chunks` takes our list of `sentences` and the `breakpoints` we found and groups the sentences into `chunks`.

Next, we need to create `embeddings` for these `chunks`:

```
# Create chunk embeddings using the create_embeddings function
chunk_embeddings = create_embeddings(text_chunks)
```

Time to generate a response:

```
Copy

# Create the user prompt based on the top chunks
user_prompt = "\n".join([f"Context {i + 1}:\n{chunk}\n=========================
user_prompt = f"{user_prompt}\nQuestion: {query}"

# Generate AI response
ai_response = generate_response(system_prompt, user_prompt)
print(ai_response.choices[0].message.content)
```

And finally, the evaluation:

```
Copy

# Create the evaluation prompt by combining the user query, AI response, true r
evaluation_prompt = f"User Query: {query}\nAI Response:\n{ai_response.choices[0

# Generate the evaluation response using the evaluation system prompt and evalu
evaluation_response = generate_response(evaluate_system_prompt, evaluation_pron

# Print the evaluation response
```

```
### OUTPUT
Based on the evaluation criteria,
I would assign a score of 0.2 to the AI assistant response.
```

> The evaluator gives this a score of just 0.2

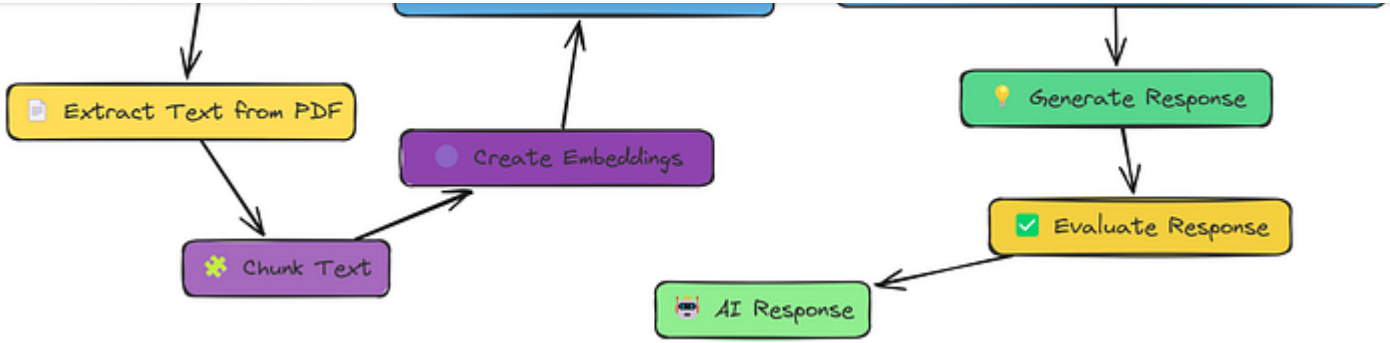
While semantic chunking *sounds* good in theory, it didn't help us here. In fact, our score went *down* compared to simple fixed-size chunking!

This shows that just changing the chunking strategy isn't a guaranteed win. We need to be more sophisticated in our approach. Let's try something else in the next section.

Context Enriched Retrieval

We saw that semantic chunking, while a good idea in principle, didn't actually improve our results.

One problem is that even semantically-defined chunks can be too focused. They might be missing crucial context from the surrounding text.



Context Enriched Workflow (Created by Fareed Khan)

Context-Enriched Retrieval addresses this by grabbing not just the best-matching chunk, but also its neighbors.

Let's see how this works in code. We'll need a new function, `context_enriched_search`, to handle the `retrieval`:

Copy

```
def context_enriched_search(query, text_chunks, embeddings, k=1, context_size=1
    """
    Retrieves the most relevant chunk along with its neighboring chunks.
    """
    # Convert the query into an embedding vector
    query_embedding = create_embeddings(query).data[0].embedding
    similarity_scores = []
```

```
        # Calculate cosine similarity between the query embedding and current c
        similarity_score = cosine_similarity(np.array(query_embedding), np.arra
        # Store the index and similarity score as a tuple
        similarity_scores.append((i, similarity_score))

    # Sort the similarity scores in descending order (highest similarity first)
    similarity_scores.sort(key=lambda x: x[1], reverse=True)

    # Get the index of the most relevant chunk
    top_index = similarity_scores[0][0]

    # Define the range for context inclusion
    # Ensure we don't go below 0 or beyond the length of text_chunks
    start = max(0, top_index - context_size)
    end = min(len(text_chunks), top_index + context_size + 1)

    # Return the relevant chunk along with its neighboring context chunks
    return [text_chunks[i] for i in range(start, end)]
```

The core logic is similar to our previous `search`, but instead of just returning the single best `chunk`, we grab a `"window"` of `chunks` around it. `context_size` controls how many `chunks` on either side we include.

Let's use this in our RAG pipeline. We'll skip the text extraction and chunking steps, as those are the same as in Simple RAG.

Now generate a response, same as before:

```
Copy

# Create the user prompt based on the top chunks
user_prompt = "\n".join([f"Context {i + 1}:\n{chunk}\n=========================
user_prompt = f"{user_prompt}\nQuestion: {query}"

# Generate AI response
ai_response = generate_response(system_prompt, user_prompt)
print(ai_response.choices[0].message.content)
```

And finally, evaluate:

```
Copy

# Create the evaluation prompt and generate the evaluation response
evaluation_prompt = f"User Query: {query}\nAI Response:\n{ai_response.choices[0
evaluation_response = generate_response(evaluate_system_prompt, evaluation_pro
print(evaluation_response.choices[0].message.content)


### OUTUT ###
```

> This time, we get an evaluation score of 0.6!

That's a significant improvement over both Simple RAG (0.5) and Semantic Chunking (0.1).

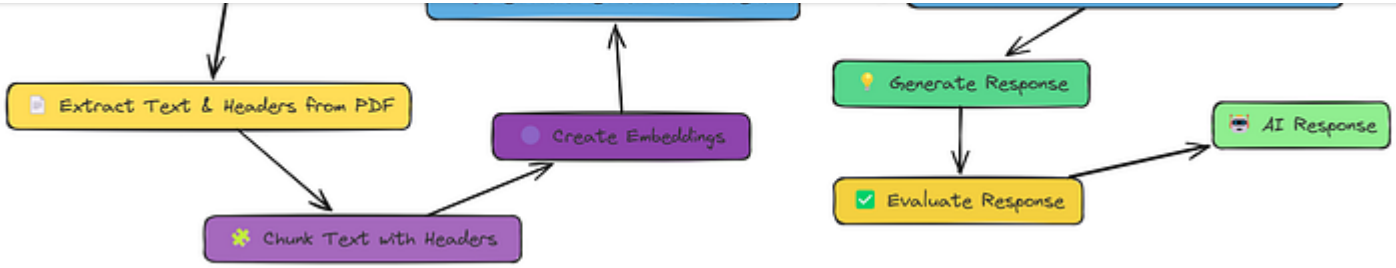
By including neighboring chunks, we've given the LLM more context to work with, and it's produced a better answer.

We're still not perfect, but we're definitely moving in the right direction. This shows how important context is for retrieval.

Contextual Chunk Headers

We've seen that adding context by including neighboring chunks helps. But what if the **content** of the chunks themselves is missing important information?

Often, documents have a clear structure titles, headings, subheadings that provide crucial context. Contextual Chunk Headers (CCH) takes advantage of this structure.



Contextual Chunk Headers (Created by Fareed Khan)

The idea is simple: before we even create our embeddings, we **prepend** a descriptive header to each chunk. This header acts like a mini-summary, giving the retrieval system (and the LLM) more to work with.

The `generate_chunk_header` function will analyze each chunk of text and generate a concise, meaningful header that summarizes its content. This helps in organizing and retrieving relevant information efficiently.

Copy

```
# Chunk the extracted text, this time generating headers
text_chunks_with_headers = chunk_text_with_headers(extracted_text, 1000, 200)

# Print a sample to see what it looks like
print("Sample Chunk with Header:")
print("Header:", text_chunks_with_headers[0]['header'])
print("Content:", text_chunks_with_headers[0]['text'])
```

```
### Output ###
Sample Chunk with Header:
Header: A Description about AI Impact
Content: AI has been an important part of society since ...
```

See how each chunk now has a header and the original text? This is the augmented data we'll use.

Now for the embeddings. We'll create embeddings for *both* the header and the text:

Copy

```
# Generate embeddings for each chunk (both header and text)
embeddings = []
for chunk in tqdm(text_chunks_with_headers, desc="Generating embeddings"):
    text_embedding = create_embeddings(chunk["text"])
    header_embedding = create_embeddings(chunk["header"])
    embeddings.append({"header": chunk["header"], "text": chunk["text"], "embed
```

We loop through our chunks, get embeddings for both the header and the text, and store everything together. This gives the retrieval system two ways to match a chunk to the query.

we perform a search, the model can consider both the high-level summary (header) and the detailed content (chunk text) to find the most relevant information.

Now, let's modify our retrieval step to return not just the matching chunks but also their headers for better context and generate response.

Copy

```
# Perform semantic search using the query and the new embeddings
top_chunks = semantic_search(query, embeddings, k=2)

# Create the user prompt based on the top chunks. note: no need to add header
# because the context is already created using header and chunk
user_prompt = "\n".join([f"Context {i + 1}:\n{chunk['text']}\n==========
user_prompt = f"{user_prompt}\nQuestion: {query}"

# Generate AI response
ai_response = generate_response(system_prompt, user_prompt)
print(ai_response.choices[0].message.content)


### OUTPUT ###
Evaluation Score: 0.5
```

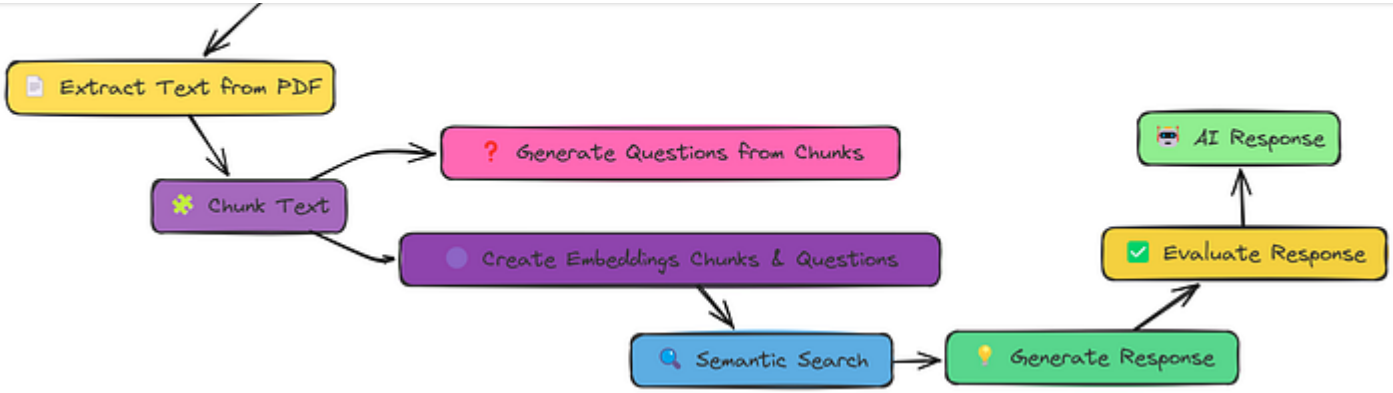
By adding those contextual headers, we've given the system a better chance to find the *right* information, and the LLM a better chance to generate a complete and accurate answer.

This shows the power of augmenting our data *before* it even goes into the retrieval system. We haven't changed the core RAG pipeline, but we've made the *data* itself more informative.

Document Augmentation

We've seen how adding context **around** our chunks (with neighbors or headers) can help. Now, let's try a different kind of augmentation: generating questions from our text chunks.

The idea is that these questions can act as alternative "queries" that might match the user's intent better than the original text chunk itself.



Document Augmentation Workflow (Created by Fareed Khan)

We add this step between the `chunking` and `embedding creation`. We can simply use `generate_questions` function for this. It takes a `text_chunk` and returns a number of questions that can be generated using it.

Let's first look at how we can achieve document augmentation with question generation:

Copy

```
# Process the document (extract text, create chunks, generate questions, build
text_chunks, vector_store = process_document(
    pdf_path,
    chunk_size=1000,
    chunk_overlap=200,
    questions_per_chunk=3
```

```
print(f"Vector store contains {len(vector_store['texts'])} items")


### OUTPUT ###
Vector store contains 214 items
```

Here, `process_document` function does it all. It takes `pdf_path`, `chunk_size`, `overlap`, and `questions_per_chunk` and returns a `vector_store`.

Now, the `vector_store` not only includes the embeddings of the document but also includes the embeddings of the generated questions.

Now, we can perform a semantic search as before, using this `vector_store`. We use a simple function here to find similar vectors.

Copy

```
# Perform semantic search to find relevant content
search_results = semantic_search(query, vector_store, k=5)

print("Query:", query)
print("\nSearch Results:")

# Organize results by type
chunk_results = []
```

```
for result in search_results:
    if result["metadata"]["type"] == "chunk":
        chunk_results.append(result)
    else:
        question_results.append(result
```

The important change here is how we handle the search results. We now have *two* types of items in our vector store: original text chunks, and generated questions. This code separates them, so we can see which type of content matched the query best.

The final steps, generating a context and then a evaluation:

Copy

```
# Prepare context from search results
context = prepare_context(search_results)

# Generate response
response_text = generate_response(query, context)

# Get reference answer from validation data
reference_answer = data[0]['ideal_answer']

# Evaluate the response
evaluation = evaluate_response(query, response_text, reference_answer)
```

```
print(evaluation)


### OUTPUT ###
Based on the evaluation criteria, I would assign a
score of 0.8 to the AI assistants response.
```

> Our evaluation shows a score of around 0.8!

Generating questions and adding *them* to our searchable index has given us another boost in performance.

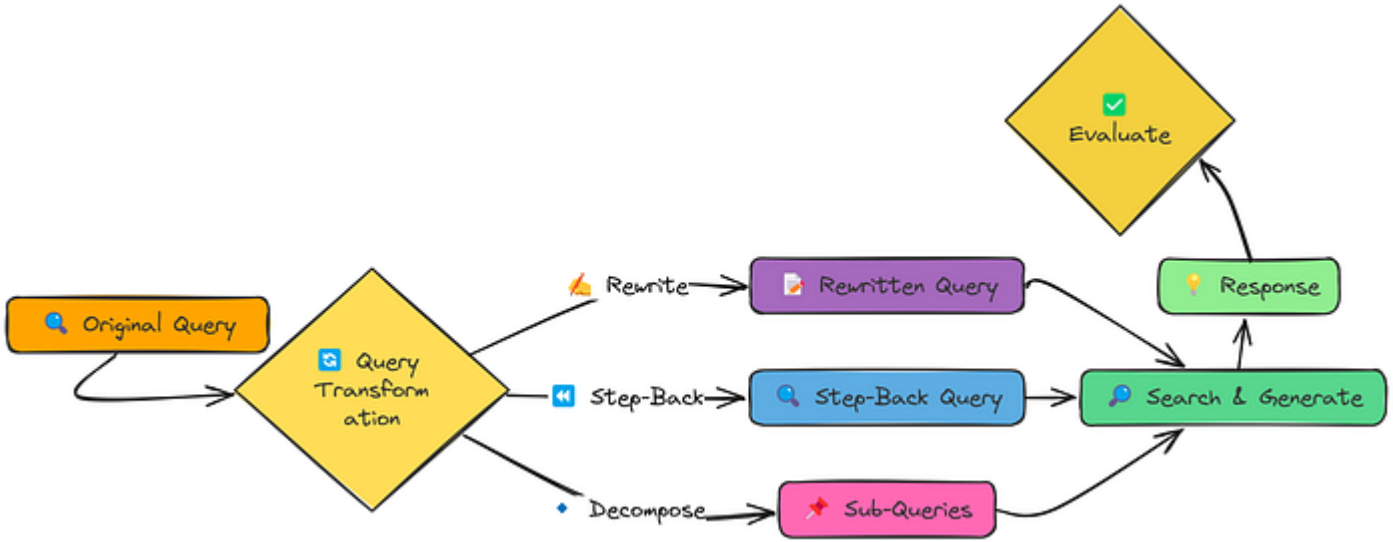
It seems that sometimes, a question is a better representation of the information need than the original text chunk.

Query Transformation

So far, we've focused on improving the data that our RAG system uses. But what about the **query** itself?

Often, the way a user phrases a question isn't the best way to search our knowledge base. Query transformations aim to fix this. We'll explore three

1. **Query Rewriting:** Making the query more specific and detailed.
2. **Step-back Prompting:** Creating a broader, more general query to retrieve background context.
3. **Sub-query Decomposition:** Breaking a complex query into multiple, simpler sub-queries.



Query Transformation Workflow (Created by Fareed Khan)

Let's see these transformations in action. We'll use our standard test query:

```
Copy
```

```
# Step-back Prompting
step_back_query = generate_step_back_query(query)
```

`generate_step_back_query` does the opposite of rewriting: it creates a broader query that might retrieve useful background information.

Finally, **sub-query decomposition:**

Copy

```
# Sub-query Decomposition
sub_queries = decompose_query(query, num_subqueries=4)
```

`decompose_query` breaks the original query into several smaller, more focused questions. The idea is that these sub-queries, taken together, might cover the original query's intent better than any single query could.

Now, to see how these transformations affect our RAG system, let's use a function that combines all previous methods:

Copy

```
    Run complete RAG pipeline with optional query transformation.

    Args:
        pdf_path (str): Path to PDF document
        query (str): User query
        transformation_type (str): Type of transformation (None, 'rewrite', 'st

    Returns:
        Dict: Results including query, transformed query, context, and response
    """
    # Process the document to create a vector store
    vector_store = process_document(pdf_path)

    # Apply query transformation and search
    if transformation_type:
        # Perform search with transformed query
        results = transformed_search(query, vector_store, transformation_type)
    else:
        # Perform regular search without transformation
        query_embedding = create_embeddings(query)
        results = vector_store.similarity_search(query_embedding, k=3)

    # Combine context from search results
    context = "\n\n".join([f"PASSAGE {i+1}:\n{result['text']}" for i, result in

    # Generate response based on the query and combined context
    response = generate_response(query, context)

    # Return the results including original query, transformation type, context
```

```
        "transformation_type": transformation_type,
        "context": context,
        "response": response
    }
```

`evaluate_transformations` function runs the original query through different query transformation techniques rewriting, step-back, and decomposition, then compares their outputs.

This helps us see which method retrieves the most relevant information for better responses.

Copy

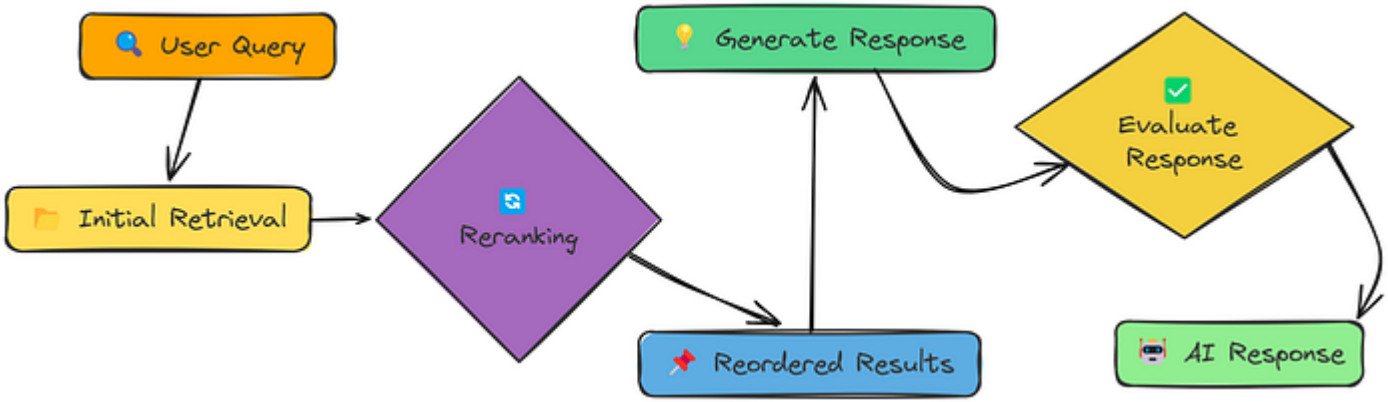
```
# Run evaluation
evaluation_results = evaluate_transformations(pdf_path, query, reference_answer
print(evaluation_results)

### OUTPUT ###
Evaluation Score: 0.5
```

The evaluation score comes in at 0.5.

While query transformations *can* be powerful, they're not a magic bullet. Sometimes, the original query is already well-formed, and trying to "improve" it can actually make things worse.

Reranker

We've tried improving the data (with chunking strategies) and the query (with transformations). Now, let's focus on the **retrieval** process itself. Simple similarity search often returns a mix of relevant and **irrelevant** results.



Reranker (Created by Fareed Khan)

The `rerank_with_llm` function takes the initial retrieved chunks and uses an LLM to reorder them based on relevance. This helps ensure that the most useful information appears first.

After reranking, a final function let's call it `generate_final_response` takes the reordered chunks, formats them into a prompt, and sends them to the LLM to generate the final response.

Copy

```
def rag_with_reranking(query, vector_store, reranking_method="llm", top_n=3, mo
    """
    Complete RAG pipeline incorporating reranking.
    """
    # Create query embedding
    query_embedding = create_embeddings(query)

    # Initial retrieval (get more than we need for reranking)
    initial_results = vector_store.similarity_search(query_embedding, k=10)

    # Apply reranking
    if reranking_method == "llm":
        reranked_results = rerank_with_llm(query, initial_results, top_n=top_n)
    elif reranking_method == "keywords":
```

```
        # No reranking, just use top results from initial retrieval
        reranked_results = initial_results[:top_n]

    # Combine context from reranked results
    context = "\n\n===\n\n".join([result["text"] for result in reranked_results]

    # Generate response based on context
    response = generate_response(query, context, model)

    return {
        "query": query,
        "reranking_method": reranking_method,
        "initial_results": initial_results[:top_n],
        "reranked_results": reranked_results,
        "context": context,
        "response": response
    }
```

It takes a `query`, a `vector_store` (which we've already created), and a `reranking_method`. We're using "llm" for LLM-based reranking. The function does the initial retrieval, calls `rerank_with_llm` to reorder the results, and then generates the response.

The `rerank_with_keywords` is defined in the notebook but I am not using it here.

```
Copy

# Run RAG with LLM-based reranking
llm_reranked_result = rag_with_reranking(query, vector_store, reranking_method=

# Evaluate.
evaluation_prompt = f"User Query: {query}\nAI Response:\n{llm_reranked_result[
evaluation_response = generate_response(evaluate_system_prompt, evaluation_pron
print(evaluation_response.choices[0].message.content)

### OUTPUT ###
Evaluation score is 0.7
```

Our evaluation score is now around 0.7!

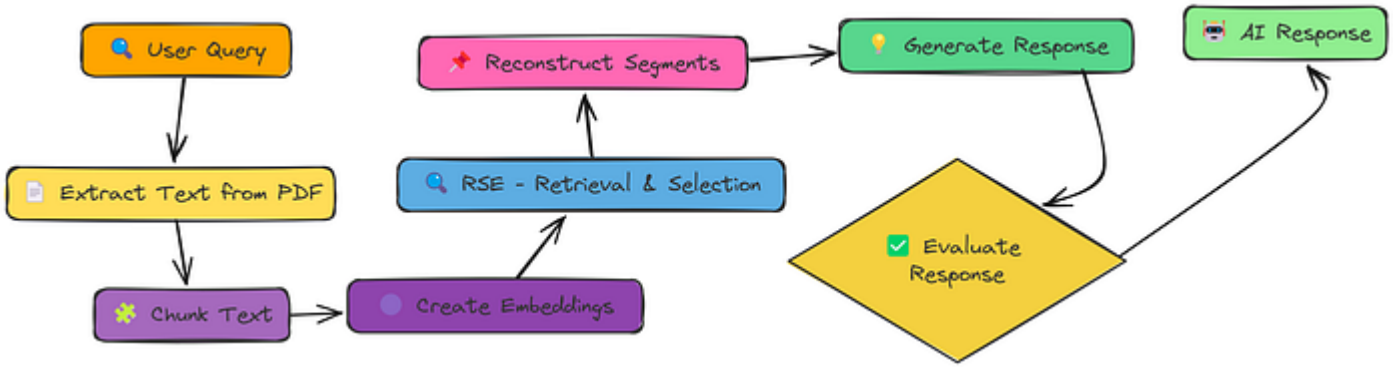
Reranking has given us a noticeable improvement. By using an LLM to directly score the relevance of each retrieved document, we're able to prioritize the *best* information for response generation.

This is a powerful technique that can significantly improve the quality of a RAG system.

RSE

addresses this.

Instead of just grabbing the top-**k** chunks, RSE tries to identify and extract entire segments of relevant text.



RSE (Created by Fareed Khan)

Let's see how we'd implement this in our existing pipeline, we are using already defined functions for `RSE`. We are adding a function call for `rag_with_rse`, it takes a `pdf_path` and `query` and returns the response. We combine several function calls to perform `RSE`.

```
Copy
```

This single line does a lot! It:

1. Processes the document (extracting text, chunking, creating embeddings, all handled *inside* rag_with_rse).
2. Calculates "chunk values" based on both relevance to the query *and* position.
3. Uses a clever algorithm to find the best *contiguous segments* of chunks.
4. Combines those segments into a context.
5. Generates a response based on that context.

Now, the evaluation:

Copy

```
# Evaluate
evaluation_prompt = f"User Query: {query}\nAI Response:\n{rse_result['response
evaluation_response = generate_response(evaluate_system_prompt, evaluation_pror
print(evaluation_response.choices[0].message.content)


### OUTPUT ###
```

> And... we've hit a score of around 0.8!

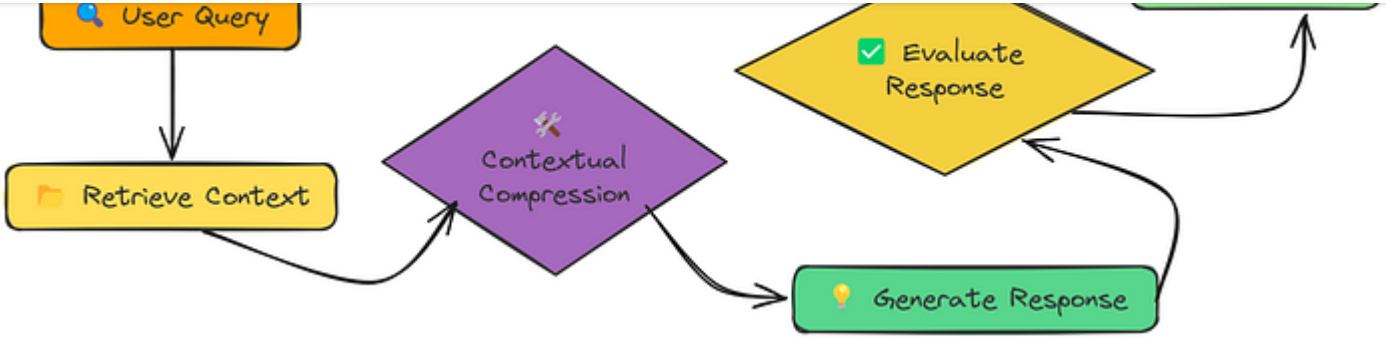
By focusing on *continuous segments* of relevant text, RSE provides the LLM with a more coherent and complete context, leading to a more accurate and comprehensive response.

This demonstrates that *how* we select and present information to the LLM is just as important as *what* information we select.

Contextual Compression

We've been adding more and more context, neighboring chunks, generated questions, entire segments. But sometimes, **less is more**.

LLMs have a limited context window, and stuffing it with irrelevant information can hurt performance.



Contextual Compression (Created by Fareed Khan)

Contextual Compression is about being selective. We retrieve a good amount of context, but then we compress it, keeping only the parts that are directly relevant to the query.

The key difference here is the **"Contextual Compression"** step before generation. We're not changing what we retrieve, but we're refining it before passing it to the LLM.

We are using here a function call `rag_with_compression`, which takes the `query` and other arguments and implements the contextual compression. Internally, it uses the LLM to analyze the retrieved chunks and extract only the sentences or paragraphs directly relevant to the `query`.

Copy

```
def rag_with_compression(pdf_path, query, k=10, compression_type="selective", r
    """
    RAG (Retrieval-Augmented Generation) pipeline with contextual compression.

    Args:
        pdf_path (str): Path to the PDF document.
        query (str): User query for retrieval.
        k (int): Number of top relevant chunks to retrieve. Default is 10.
        compression_type (str): Type of compression to apply to retrieved chunk
        model (str): Language model to use for response generation. Default is

    Returns:
        dict: A dictionary containing the query, original and compressed chunks
    """

    print(f"\n=== RAG WITH COMPRESSION ===\nQuery: {query} | Compression: {comp

    # Process the document to extract, chunk, and embed text
    vector_store = process_document(pdf_path)

    # Retrieve top-k relevant chunks based on query similarity
    results = vector_store.similarity_search(create_embeddings(query), k=k)
    retrieved_chunks = [r["text"] for r in results]

    # Apply compression to retrieved chunks
    compressed = batch_compress_chunks(retrieved_chunks, query, compression_typ
```

```
    # Combine compressed chunks to form context for response generation
    context = "\n\n---\n\n".join(compressed_chunks)

    # Generate a response using the compressed context
    response = generate_response(query, context, model)

    print(f"\n=== RESPONSE ===\n{response}")

    # Return detailed results
    return {
        "query": query,
        "original_chunks": retrieved_chunks,
        "compressed_chunks": compressed_chunks,
        "compression_ratios": compression_ratios,
        "context_length_reduction": f"{sum(compression_ratios)/len(compression_
        "response": response
    }
```

rag_with_compression gives options for different compression types:

- **"selective"** Keeps only the *directly* relevant sentences.
- **"summary"** Creates a short summary focused on the query.
- **"extraction"** Extracts *only* the sentences that contain the answer (very strict!).

Now, to run the compression we use this code:

```
# Run RAG with contextual compression (using "selective" mode)
compression_result = rag_with_compression(pdf_path, query, compression_type="se

# Evaluate.
evaluation_prompt = f"User Query: {query}\nAI Response:\n{compression_result['r
evaluation_response = generate_response(evaluate_system_prompt, evaluation_pron
print(evaluation_response.choices[0].message.content)


### OUTPUT ###
Evaluation Score 0.75
```

> This give us a score around 0.75.

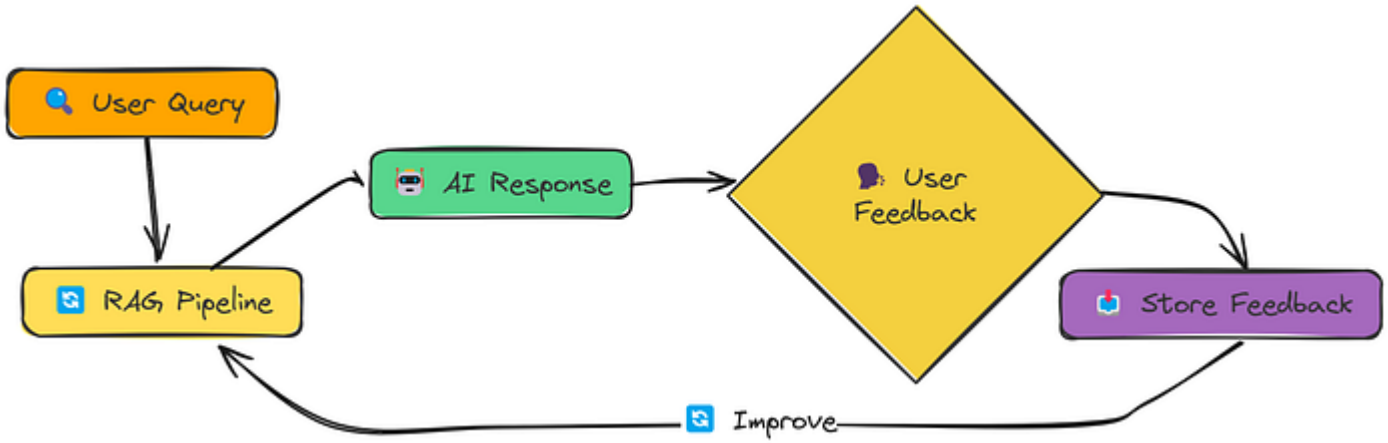
Contextual compression is a powerful technique because it balances *breadth* (initial retrieval gets a wide range of information) and *focus* (compression removes the noise).

By giving the LLM only the *most* relevant information, we often get more concise and accurate answers.

Feedback Loop

The idea is simple:

1. The user provides feedback on the RAG system's response (e.g., good/bad, relevant/irrelevant).
2. The system stores this feedback.
3. Future retrievals use this feedback to improve.



Feedback Loop (Created by Fareed Khan)

We can implement a feedback loop using a function call `full_rag_workflow`. Here is the function definition.

```
def full_rag_workflow(pdf_path, query, feedback_data=None, feedback_file="feedb
    """
    Execute a complete RAG workflow with feedback integration for continuous im

    """
    # Step 1: Load historical feedback for relevance adjustment if not explicit
    if feedback_data is None:
        feedback_data = load_feedback_data(feedback_file)
        print(f"Loaded {len(feedback_data)} feedback entries from {feedback_fil

    # Step 2: Process document through extraction, chunking and embedding pipel
    chunks, vector_store = process_document(pdf_path)

    # Step 3: Fine-tune the vector index by incorporating high-quality past int
    # This creates enhanced retrievable content from successful Q&A pairs
    if fine_tune and feedback_data:
        vector_store = fine_tune_index(vector_store, chunks, feedback_data)

    # Step 4: Execute core RAG with feedback-aware retrieval
    # Note: This depends on the rag_with_feedback_loop function which should be
    result = rag_with_feedback_loop(query, vector_store, feedback_data)

    # Step 5: Collect user feedback to improve future performance
    print("\n=== Would you like to provide feedback on this response? ===")
    print("Rate relevance (1-5, with 5 being most relevant):")
    relevance = input()

    print("Rate quality (1-5, with 5 being highest quality):")
    quality = input()
```

```
    comments = input()

    # Step 6: Format feedback into structured data
    feedback = get_user_feedback(
        query=query,
        response=result["response"],
        relevance=int(relevance),
        quality=int(quality),
        comments=comments
    )

    # Step 7: Persist feedback to enable continuous system learning
    store_feedback(feedback, feedback_file)
    print("Feedback recorded. Thank you!")

    return result
```

This full_rag_workflow function does several things:

1. **Loads existing feedback:** It checks for a feedback_data.json file and loads any previous feedback.
2. **Runs the RAG pipeline:** This part is similar to what we've done before.
3. **Asks for feedback:** It prompts the user to rate the relevance and quality of the response.
4. **Stores the feedback:** It saves the feedback to the feedback_data.json file.

`adjust_relevance_scores` (which are not shown here for brevity). But the key idea is that good feedback can boost the relevance of certain documents, and bad feedback can lower it.

Let's run a simplified version, assuming we don't have any existing feedback:

```
Copy

# we don't have previous feedback, therefore "fine_tune=False"
result = full_rag_workflow(pdf_path=pdf_path, query=query, fine_tune=False)

# Evaluate.
evaluation_prompt = f"User Query: {query}\nAI Response:\n{result['response']}\r
evaluation_response = generate_response(evaluate_system_prompt, evaluation_pror
print(evaluation_response.choices[0].message.content)


### OUTPUT ###
Evaluation score is 0.7 because ....
```

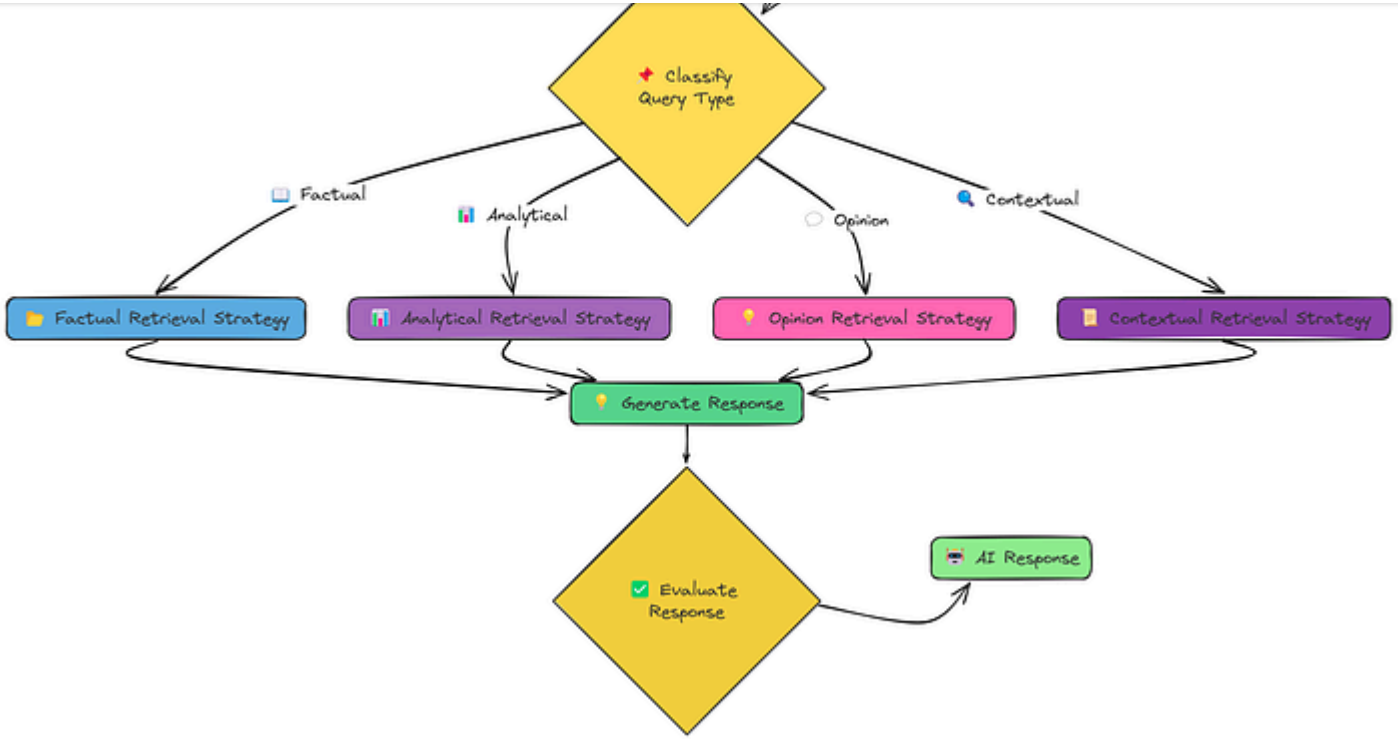
> We see a score around 0.7!

It's not a *huge* jump, and that's expected. A feedback loop improves the system *over time*, with repeated interactions. This section just demonstrates the

The real power comes from accumulating feedback and using it to refine the retrieval process. This makes the RAG system *adaptive* and *personalized* to the kinds of queries it receives.

Adaptive RAG

We've explored various ways to improve RAG: better chunking, adding context, transforming queries, reranking, and even incorporating feedback.



Adaptive Rag Workflow (Created by Fareed Khan)

But what if the best technique depends on the type of question being asked? That's the idea behind Adaptive RAG.

We are using here four different strategies:

1. **Factual Strategy:** Focuses on retrieving precise facts and figures.

3. **Opinion Strategy:** Tries to gather diverse viewpoints on a subjective issue.
4. **Contextual Strategy:** Incorporates user-specific context to tailor the retrieval.

Let's see how this works. We'll use a function called `rag_with_adaptive_retrieval` to handle the entire process:

Copy

```
def rag_with_adaptive_retrieval(pdf_path, query, k=4, user_context=None):
    """
    Complete RAG pipeline with adaptive retrieval.

    """
    print("\n=== RAG WITH ADAPTIVE RETRIEVAL ===")
    print(f"Query: {query}")

    # Process the document to extract text, chunk it, and create embeddings
    chunks, vector_store = process_document(pdf_path)

    # Classify the query to determine its type
    query_type = classify_query(query)
    print(f"Query classified as: {query_type}")

    # Retrieve documents using the adaptive retrieval strategy based on the que
    retrieved_docs = adaptive_retrieval(query, vector_store, k, user_context)
```

```
    response = generate_response(query, retrieved_docs, query_type)

    # Compile the results into a dictionary
    result = {
        "query": query,
        "query_type": query_type,
        "retrieved_documents": retrieved_docs,
        "response": response
    }

    print("\n=== RESPONSE ===")
    print(response)

    return result
```

It first classifies the query using a function called `classify_query` that is defined with other helper functions.

Based on the identified type, it selects and executes the appropriate specialized retrieval strategy (`factual_retrieval_strategy` , `analytical_retrieval_strategy` , `opinion_retrieval_strategy` , or `contextual_retrieval_strategy`). Finally, it uses `generate_response` to generate the response using the retrieved documents.

The function returns a dictionary containing the results, including the `query` , `query type` , `retrieved documents` , and the `generated response` .

```
Copy

# Run the adaptive RAG pipeline
result = rag_with_adaptive_retrieval(pdf_path, query)


# Evaluate.
evaluation_prompt = f"User Query: {query}\nAI Response:\n{result['response']}\r
evaluation_response = generate_response(evaluate_system_prompt, evaluation_pror
print(evaluation_response.choices[0].message.content)


### OUTPUT ###
Evaluation score is 0.86
```

We have achieved a score of around 0.856 this time.

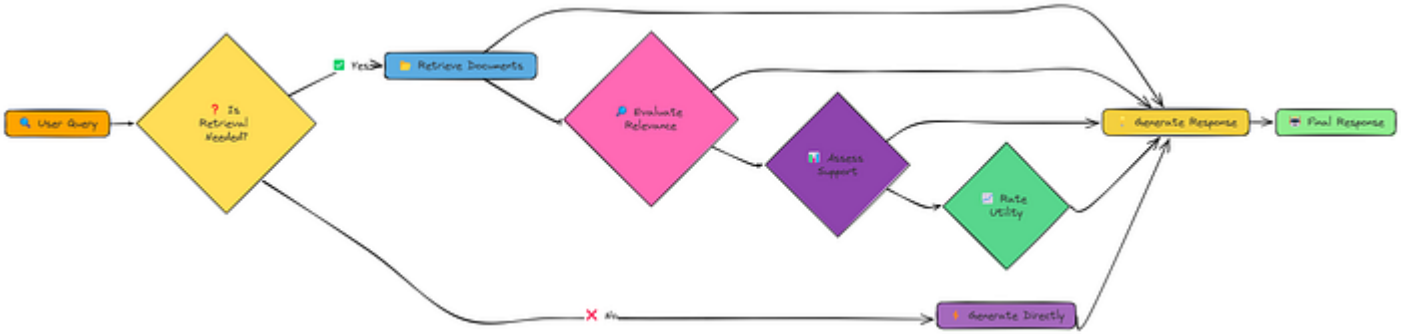
By adapting our retrieval strategy to the specific type of query, we can achieve significantly better results than with a one-size-fits-all approach. This highlights the importance of understanding the user's intent and tailoring the RAG system accordingly.

The Adaptive RAG is not a fixed procedure, it is a framework that give use the functionality to select best strategies base on the query.

Self RAG

Up to this point, our RAG systems have been largely **reactive**. They take a query, retrieve information, and generate a response. Self-RAG takes a different approach: it's **proactive** and **reflective**.

It doesn't just retrieve and generate, it **thinks** about **whether** to retrieve, **what** to retrieve, and **how** to use the retrieved information.



Self RAG Workflow (Created by Fareed Khan)

These **"reflection"** steps allow Self-RAG to be much more dynamic and adaptable than traditional RAG. It can decide to:

- Skip retrieval entirely.
- Retrieve multiple times with different strategies.

- Prioritize well-supported and useful information.

The core of Self-RAG lies in its ability to generate "reflection tokens". These are special tokens that the model uses to *reason* about its own process. For example it uses different tokens for, **retrieval_needed**, **relevance**, **support_rating** and **utility_ratings**.

The model uses the combination of these tokens to decide when it has to retrieve and when not, and on what bases the LLM should generate the final response.

First, deciding whether retrieval is needed:

Copy

```
def determine_if_retrieval_needed(query):
    """
    (Illustrative Example - NOT fully functional)
    Determines if retrieval is necessary for the given query.
    """
    system_prompt = """You are an AI assistant that determines if retrieval is
    For factual questions, specific information requests, or questions about ev
    For opinions, hypothetical scenarios, or simple queries with common knowled
    Answer with ONLY "Yes" or "No"."""

    user_prompt = f"Query: {query}\n\nIs retrieval necessary to answer this que
```

```
        model="meta-llama/Llama-3.2-3B-Instruct",
        messages=[
            {"role": "system", "content": system_prompt},
            {"role": "user", "content": user_prompt}
        ],
        temperature=0
    )
    answer = response.choices[0].message.content.strip().lower()
    return "yes" in answer
```

This `determine_if_retrieval_needed` function (again, simplified) uses an LLM to make a judgment call about whether external information is needed.

- For a factual question like **"What is the capital of France?"**, it might return `False` (the LLM likely already knows this).
- For a creative task like **"Write a poem..."**, it would also likely return `False`.
- But for a more complex or niche query, it would return `True`.

Here's a simplified example of relevance evaluation:

Copy

```
def evaluate_relevance(query, context):
    """
    (Illustrative Example - NOT fully functional)
```

```
system_prompt = """You are an AI assistant. Determine if a document is rel
Answer with ONLY "Relevant" or "Irrelevant"."""

user_prompt = f"""Query: {query}
Document content:
{context[:500]}... [truncated]

Is this document relevant to the query? Answer with ONLY "Relevant" or "Irr
"""

response = client.chat.completions.create(
    model="meta-llama/Llama-3.2-3B-Instruct",
    messages=[
        {"role": "system", "content": system_prompt},
        {"role": "user", "content": user_prompt}
    ],
    temperature=0
)
answer = response.choices[0].message.content.strip().lower()
return answer
```

This `evaluate_relevance` function (again, simplified) uses an LLM to judge whether a retrieved document is relevant to the `query`.

This allows **Self-RAG** to filter out irrelevant documents before generating a response.

```
Copy

# we can call `self_rag` function for self-rag, and it automatically
# decide when to retrieve and when not.
result = self_rag(query, vector_store)

print(result["response"])


### OUTPUT ###
Evaluation score for the AI Response is 0.65
```

We got a score of 0.6 here.

This reflects the fact that:

- Self-RAG has great *potential*, but a full implementation is complex.
- Even the "Is Retrieval Needed?" step, which we showed, can be wrong sometimes.
- We haven't shown the full "reflection" process, so we can't claim a higher score.

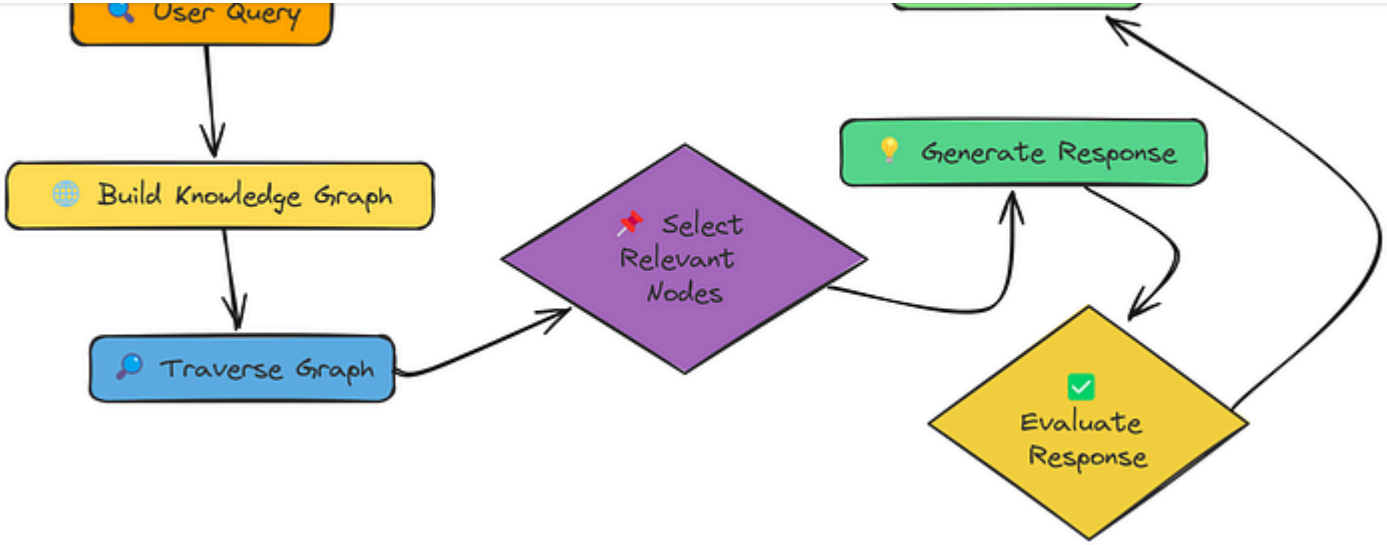
knowledge and retrieval needs.

Knowledge Graph

So far, our RAG systems have treated documents as collections of independent chunks. But what if the information is connected? What if understanding one concept requires understanding related concepts? That's where Graph RAG comes in.

Instead of a flat list of chunks, Graph RAG organizes information as a knowledge graph. Think of it like a network:

1. **Nodes:** Represent concepts, entities, or pieces of information (like our text chunks).
2. **Edges:** Represent relationships between those nodes.



Knowledge Graph Workflow (Created by Fareed Khan)

The core idea is that by traversing this graph, we can find not just directly relevant information, but also *indirectly* relevant information that provides crucial context.

Let's see some simplified code of how the core steps work: First, build the knowledge graph:

Copy

```
def build_knowledge_graph(chunks):
    """
```

```
    Args:
        chunks (list of dict): List of text chunks, each containing a "text" fi

    Returns:
        tuple: (Graph with nodes as text chunks, list of embeddings)
    """
    graph, texts = nx.Graph(), [c["text"] for c in chunks]
    embeddings = create_embeddings(texts)  # Compute embeddings

    # Add nodes with extracted concepts and embeddings
    for i, (chunk, emb) in enumerate(zip(chunks, embeddings)):
        graph.add_node(i, text=chunk["text"], concepts := extract_concepts(chur

    # Create edges based on shared concepts and embedding similarity
    for i, j in ((i, j) for i in range(len(chunks)) for j in range(i + 1, len(c
        if shared_concepts := set(graph.nodes[i]["concepts"]) & set(graph.nodes
            sim = np.dot(embeddings[i], embeddings[j]) / (np.linalg.norm(embedc
            weight = 0.7 * sim + 0.3 * (len(shared_concepts) / min(len(graph.nc
            if weight > 0.6:
                graph.add_edge(i, j, weight=weight, similarity=sim, shared_conc

    print(f"Graph built: {graph.number_of_nodes()} nodes, {graph.number_of_edge
    return graph, embeddings
```

It takes a `query`, a `graph`, and `embeddings`, and returns a list of relevant nodes and a traversal path.

Copy

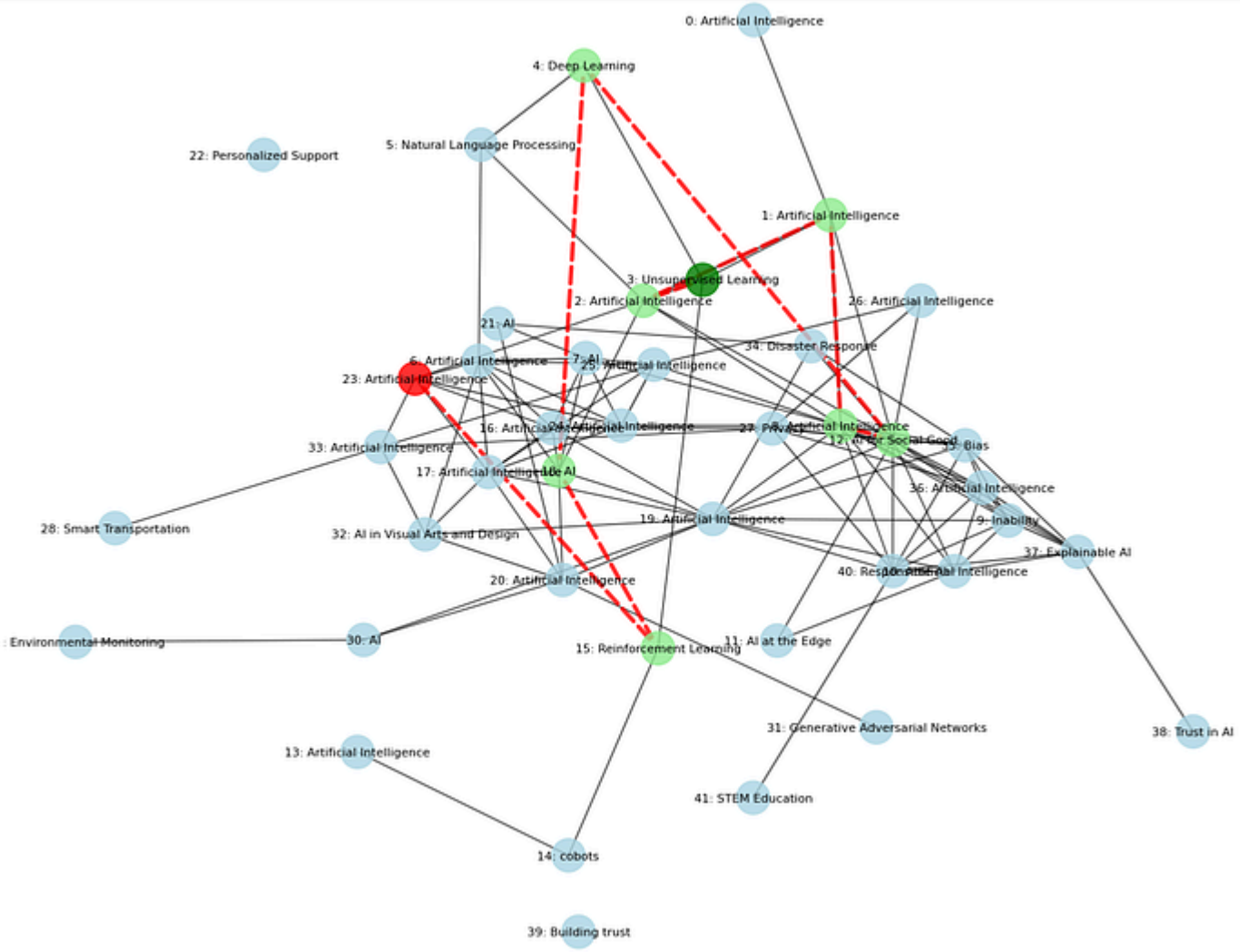
```
def graph_rag_pipeline(pdf_path, query, chunk_size=1000, chunk_overlap=200, top
    """
    Complete Graph RAG pipeline from document to answer.
    """
    # Extract text from the PDF document
    text = extract_text_from_pdf(pdf_path)

    # Split the extracted text into overlapping chunks
    chunks = chunk_text(text, chunk_size, chunk_overlap)

    # Build a knowledge graph from the text chunks
    graph, embeddings = build_knowledge_graph(chunks)

    # Traverse the knowledge graph to find relevant information for the query
    relevant_chunks, traversal_path = traverse_graph(query, graph, embeddings,

    # Generate a response based on the query and the relevant chunks
    response = generate_response(query, relevant_chunks)

    # Return the query, response, relevant chunks, traversal path, and the grap
    return {
        "query": query,
        "response": response,
        "relevant_chunks": relevant_chunks,
        "traversal_path": traversal_path,
```

Let's use this to generate a response:

```
Copy

# Execute the Graph RAG pipeline to process the document and answer the query
results = graph_rag_pipeline(pdf_path, query)

# Evaluate.
evaluation_prompt = f"User Query: {query}\nAI Response:\n{results['response']}\
evaluation_response = generate_response(evaluate_system_prompt, evaluation_pror
print(evaluation_response.choices[0].message.content)


### OUTPUT
0.78
```

> We got a score of around 0.78.

Graph RAG didn't outperforms simpler methods but it can capture the *relationships* between pieces of information, not just the individual pieces themselves.



KG on our validation pdf (Created by Fareed Khan)

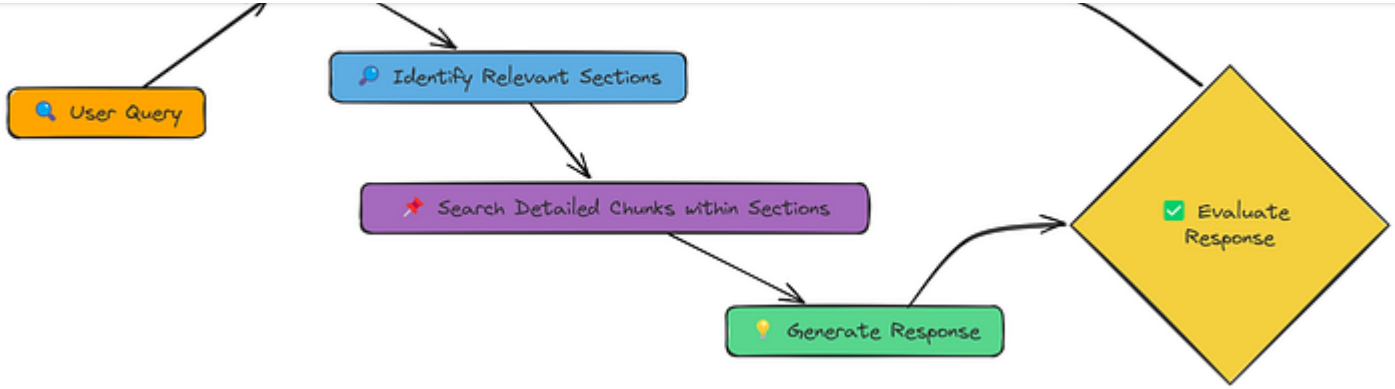
Hierarchical Indices

We've explored various ways to improve RAG: better chunking, context enrichment, query transformations, reranking, and even graph-based retrieval. But there's a fundamental trade-off:

- Small chunks: Good for precise matching, but lose context.
- Large chunks: Preserve context, but can lead to less relevant retrievals.

Hierarchical Indices offer a solution: we create two levels of representation:

1. Summaries: Concise overviews of larger sections of the document.
2. Detailed Chunks: Smaller chunks within those sections.



Hierarchical Indices Workflow (Created by Fareed Khan)

1. First, **search the summaries:** This quickly narrows down the relevant sections of the document.
2. Then, **search the detailed chunks only within those sections:** This provides the precision of small chunks while maintaining the context of the larger section.

Let's see this in action using a function call `hierarchical_rag`:

Copy

```
def hierarchical_rag(query, pdf_path, chunk_size=1000, chunk_overlap=200,
                     k_summaries=3, k_chunks=5, regenerate=False):
    """
    Complete hierarchical Retrieval-Augmented Generation (RAG) pipeline.
```

```
    pdf_path (str): Path to the PDF document.
    chunk_size (int): Size of text chunks for processing.
    chunk_overlap (int): Overlap between consecutive chunks.
    k_summaries (int): Number of top summaries to retrieve.
    k_chunks (int): Number of detailed chunks to retrieve per summary.
    regenerate (bool): Whether to reprocess the document.

Returns:
    dict: Contains the query, generated response, retrieved chunks,
          and counts of summaries and detailed chunks.
"""
# Define filenames for caching summary and detailed vector stores
summary_store_file = f"{os.path.basename(pdf_path)}_summary_store.pkl"
detailed_store_file = f"{os.path.basename(pdf_path)}_detailed_store.pkl"

# Process document if regeneration is required or cache files are missing
if regenerate or not os.path.exists(summary_store_file) or not os.path.exis
    print("Processing document and creating vector stores...")
    summary_store, detailed_store = process_document_hierarchically(pdf_pat

    # Save processed stores for future use
    with open(summary_store_file, 'wb') as f:
        pickle.dump(summary_store, f)
    with open(detailed_store_file, 'wb') as f:
        pickle.dump(detailed_store, f)
else:
    # Load existing vector stores from cache
    print("Loading existing vector stores...")
    with open(summary_store_file, 'rb') as f:
```

```
        detailed_store = pickle.load(f)

    # Retrieve relevant chunks using hierarchical search
    retrieved_chunks = retrieve_hierarchically(query, summary_store, detailed_s

    # Generate a response based on the retrieved chunks
    response = generate_response(query, retrieved_chunks)

    # Return results with metadata
    return {
        "query": query,
        "response": response,
        "retrieved_chunks": retrieved_chunks,
        "summary_count": len(summary_store.texts),
        "detailed_count": len(detailed_store.texts)
    }
```

This `hierarchical_rag` function handles the **two-stage retrieval** process:

1. First, it searches the `summary_store` to find the most relevant summaries.
2. Then, it searches the `detailed_store`, but only within the chunks belonging to the top summaries. This is much more efficient than searching all the detailed chunks.

The function also has a `regenerate` argument to create a new vector store or use the existing one.

```
Copy

# Run the hierarchical RAG pipeline
result = hierarchical_rag(query, pdf_path)
```

We retrieve and generate the response. Finally, let see the evaluation score:

```
Copy

# Evaluate.
evaluation_prompt = f"User Query: {query}\nAI Response:\n{result['response']}\r
evaluation_response = generate_response(evaluate_system_prompt, evaluation_pror
print(evaluation_response.choices[0].message.content)


### OUTPUT
0.84
```

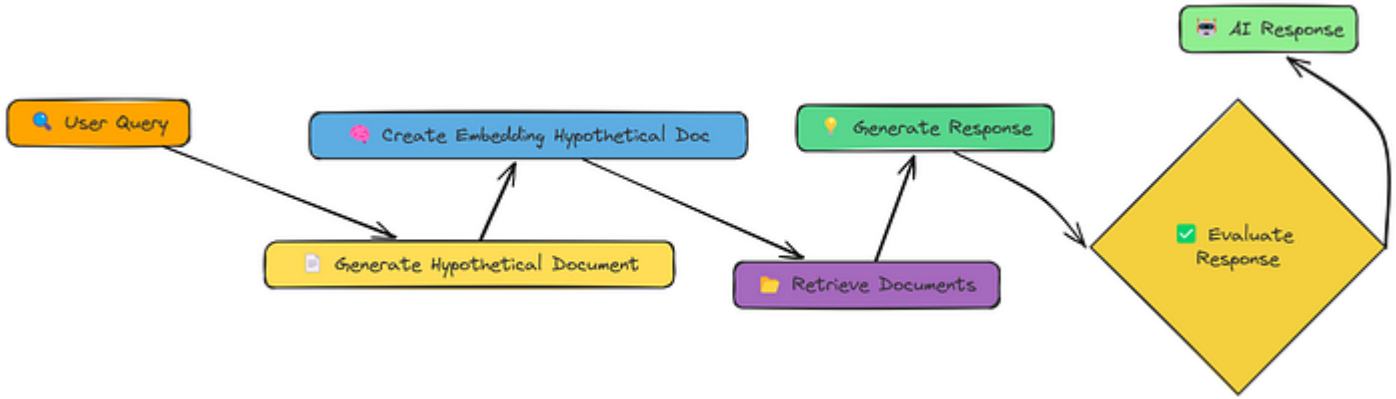
Our score is 0.84 😆

Hierarchical retrieval provides the best score yet.

We get the *speed* of searching summaries, and the *precision* of searching smaller chunks, *plus* the added context that comes from knowing *which section* each

HyDE

So far, we've been directly embedding either the user's query or transformed versions of it. HyDE (Hypothetical Document Embedding) takes a different approach. Instead of embedding the query, it embeds a hypothetical document that answers the query.



HyDE Workflow (Created by Fareed Khan)

The flow is:

1. **Generate a hypothetical document:** Use the LLM to create a document that *would* answer the query, if it existed.

3. **Retrieve:** Find documents similar to the *hypothetical document's* embedding.
4. **Generate:** Use the *retrieved* documents (not the hypothetical one!) to answer the query.

The idea is that a full document, even a hypothetical one, is a richer semantic representation than a short query. This can help bridge the gap between the query and the documents in the embedding space.

Let's see how this works. First, we need a function to generate that hypothetical document.

We use `generate_hypothetical_document` to do so:

Copy

```
def generate_hypothetical_document(query, desired_length=1000):
    """
    Generate a hypothetical document that answers the query.
    """
    # Define the system prompt to instruct the model on how to generate the doc
    system_prompt = f"""You are an expert document creator.
    Given a question, generate a detailed document that would directly answer t
    The document should be approximately {desired_length} characters long and p
```

```
    Do not mention that this is a hypothetical document - just write the conte

    # Define the user prompt with the query
    user_prompt = f"Question: {query}\n\nGenerate a document that fully answers

    # Make a request to the OpenAI API to generate the hypothetical document
    response = client.chat.completions.create(
        model="meta-llama/Llama-3.2-3B-Instruct",  # Specify the model to use
        messages=[
            {"role": "system", "content": system_prompt},  # System message to
            {"role": "user", "content": user_prompt}  # User message with the c
        ],
        temperature=0.1  # Set the temperature for response generation
    )

    # Return the generated document content
    return response.choices[0].message.content
```

This function takes the query and uses an LLM to *invent* a document that answers it.

Now, let's put it all together in a hyde_rag function:

Copy

```
def hyde_rag(query, vector_store, k=5, should_generate_response=True):
    """
```

```
print(f"\n=== Processing query with HyDE: {query} ===\n")

# Step 1: Generate a hypothetical document that answers the query
print("Generating hypothetical document...")
hypothetical_doc = generate_hypothetical_document(query)
print(f"Generated hypothetical document of {len(hypothetical_doc)} characte

# Step 2: Create embedding for the hypothetical document
print("Creating embedding for hypothetical document...")
hypothetical_embedding = create_embeddings([hypothetical_doc])[0]

# Step 3: Retrieve similar chunks based on the hypothetical document
print(f"Retrieving {k} most similar chunks...")
retrieved_chunks = vector_store.similarity_search(hypothetical_embedding, k

# Prepare the results dictionary
results = {
    "query": query,
    "hypothetical_document": hypothetical_doc,
    "retrieved_chunks": retrieved_chunks
}

# Step 4: Generate a response if requested
if should_generate_response:
    print("Generating final response...")
    response = generate_response(query, retrieved_chunks)
    results["response"] = response
```

The hyde_rag function now:

1. Generates the hypothetical document.
2. Creates an embedding of that *document* (not the query!).
3. Uses that embedding for retrieval.
4. Generates a response, as before.

Let's run it and see the generated response:

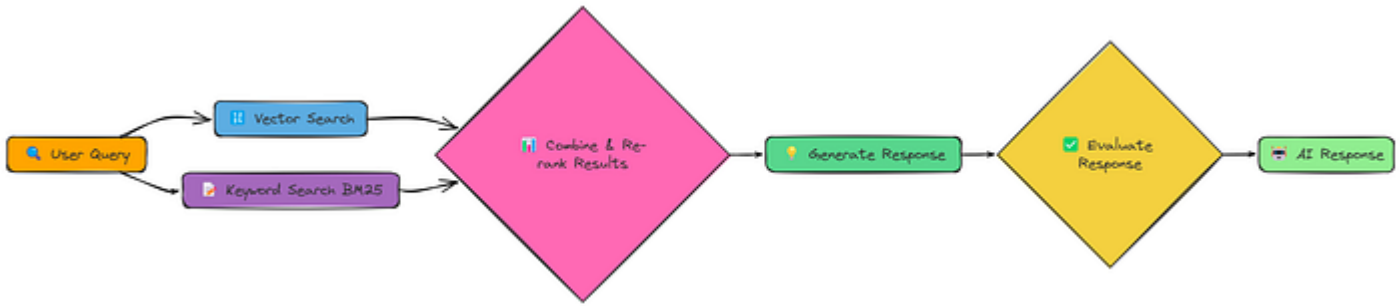
```
Copy

# Run HyDE RAG
hyde_result = hyde_rag(query, vector_store)

# Evaluate.
evaluation_prompt = f"User Query: {query}\nAI Response:\n{hyde_result['response
evaluation_response = generate_response(evaluate_system_prompt, evaluation_pro
print(evaluation_response.choices[0].message.content)


### OUTPUT
0.5
```

While HyDE is a clever idea, it doesn't *always* work better. In this case, the hypothetical document might have gone in a slightly different direction than our actual document collection, leading to less relevant retrievals.

The key lesson here is that there's no single "best" RAG technique. Different approaches work better for different queries and different data.

Fusion

We've seen that different retrieval methods have different strengths. Vector search is good at semantic similarity, while keyword search excels at finding exact matches. What if we could combine them? That's the idea behind Fusion RAG.



Fusion RAG workflow (Created by Fareed Khan)

precise keyword matches.

The core of our implementation is the fusion_retrieval function. This function performs both vector-based and BM25-based retrieval, normalizes the scores from each, combines them using a weighted formula and then, ranks documents according to the combined score.

Here is the function of fusion retrieval:

Copy

```
import numpy as np

def fusion_retrieval(query, chunks, vector_store, bm25_index, k=5, alpha=0.5):
    """Perform fusion retrieval by combining vector-based and BM25 search resul

    # Generate embedding for the query
    query_embedding = create_embeddings(query)

    # Perform vector search and store results in a dictionary (index -> similar
    vector_results = {
        r["metadata"]["index"]: r["similarity"]
        for r in vector_store.similarity_search_with_scores(query_embedding, le
    }
```

```
    for r in bm25_search(bm25_index, chunks, query, len(chunks))
}

# Retrieve all documents from the vector store
all_docs = vector_store.get_all_documents()

# Compute combined scores for each document using a weighted sum of vector
scores = [
    (i, alpha * vector_results.get(i, 0) + (1 - alpha) * bm25_results.get(i
    for i in range(len(all_docs))
]

# Sort documents by combined score in descending order and keep the top k r
top_docs = sorted(scores, key=lambda x: x[1], reverse=True)[:k]

# Return the top k documents with text, metadata, and combined score
return [
    {"text": all_docs[i]["text"], "metadata": all_docs[i]["metadata"], "sco
    for i, s in top_docs
]
```

It takes the best of both approaches:

- **Vector Search:** Uses our existing create_embeddings and SimpleVectorStore for semantic similarity.

- **Score Combination:** Combines the scores from both methods, giving us a single, unified ranking.

Let's run the complete pipeline and generate response:

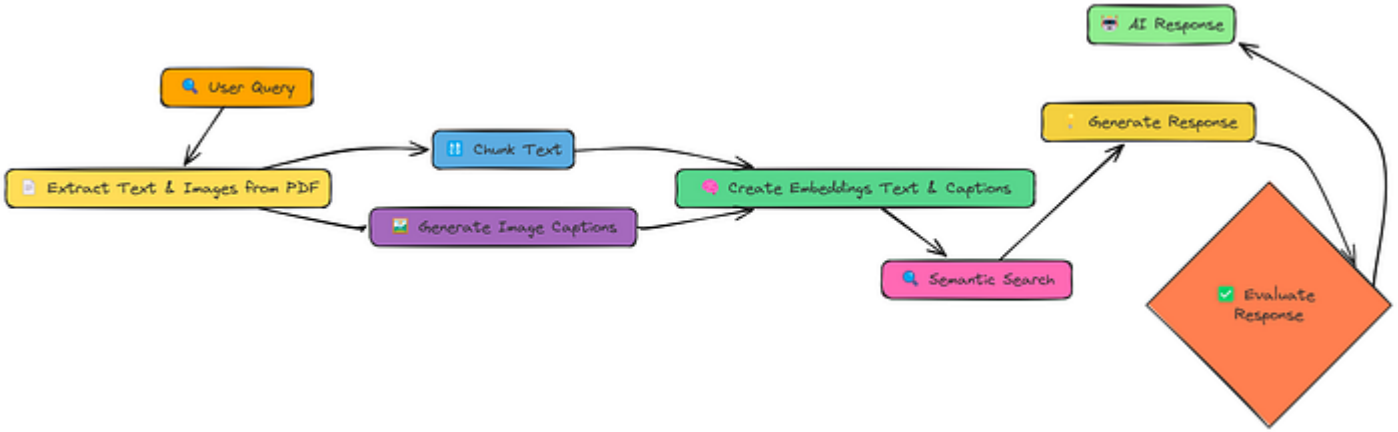
```
Copy

# First, process the document to create chunks, vector store, and BM25 index
chunks, vector_store, bm25_index = process_document(pdf_path)

# Run RAG with fusion retrieval
fusion_result = answer_with_fusion_rag(query, chunks, vector_store, bm25_index)
print(fusion_result["response"])

# Evaluate.
evaluation_prompt = f"User Query: {query}\nAI Response:\n{fusion_result['respor
evaluation_response = generate_response(evaluate_system_prompt, evaluation_pror
print(evaluation_response.choices[0].message.content)


### OUTPUT
Evaluation score for AI Response is 0.83
```

Final score is 0.83.

It's like having two experts working together one good at understanding the *meaning* of the query, and the other good at finding *exact matches*.

Multi Model

Up until now, we've only dealt with text. But a lot of information is locked up in images, charts, and diagrams. Multi-Modal RAG aims to unlock that information and use it to improve our responses.



Multi Model Workflow (Created by Fareed Khan)

The key changes here are:

2. **Generate Image Captions:** We use an LLM (specifically, a model with *vision* capabilities) to generate text descriptions (captions) for each image.
3. **Create Embeddings (Text & Captions):** We create embeddings for *both* the text chunks *and* the image captions.
4. **Embeddings model**: In this notebook we are using BAAI/bge-en-icl embedding model.
5. **LLM model**: For generating the response and image caption we will use llava-hf/llava-1.5–7b-hf model.

This way, our vector store contains both textual and visual information, and we can search across *both* modalities.

Here we define the `process_document` function:

Copy

```
def process_document(pdf_path, chunk_size=1000, chunk_overlap=200):
    """
    Process a document for multi-modal RAG.

    """
    # Create a directory for extracted images
    image_dir = "extracted_images"
```

```
# Extract text and images from the PDF
text_data, image_paths = extract_content_from_pdf(pdf_path, image_dir)

# Chunk the extracted text
chunked_text = chunk_text(text_data, chunk_size, chunk_overlap)

# Process the extracted images to generate captions
image_data = process_images(image_paths)

# Combine all content items (text chunks and image captions)
all_items = chunked_text + image_data

# Extract content for embedding
contents = [item["content"] for item in all_items]

# Create embeddings for all content
print("Creating embeddings for all content...")
embeddings = create_embeddings(contents)

# Build the vector store and add items with their embeddings
vector_store = MultiModalVectorStore()
vector_store.add_items(all_items, embeddings)

# Prepare document info with counts of text chunks and image captions
doc_info = {
    "text_count": len(chunked_text),
    "image_count": len(image_data),
    "total_items": len(all_items),
}
```

```
    print(f"Added {len(all_items)} items to vector store ({len(chunked_text)} t

    # Return the vector store and document info
    return vector_store, doc_info
```

This function handles the **image extraction and captioning**, and the creation of a `MultiModalVectorStore`.

We're making the assumption that **image captioning** works reasonably well. *(In a real-world scenario, you'd want to carefully evaluate the quality of your captions).*

Now, let's put it all together with a query:

Copy

```
# Process the document to create vector store. we have a new pdf for this
pdf_path = "data/attention_is_all_you_need.pdf"
vector_store, doc_info = process_document(pdf_path)

# Run the multi-modal RAG pipeline.  This is very similar to before!
result = query_multimodal_rag(query, vector_store)

# Evaluate.
evaluation_prompt = f"User Query: {query}\nAI Response:\n{result['response']}\n
evaluation_response = generate_response(evaluate_system_prompt, evaluation_pro
print(evaluation_response.choices[0].message.content)
```

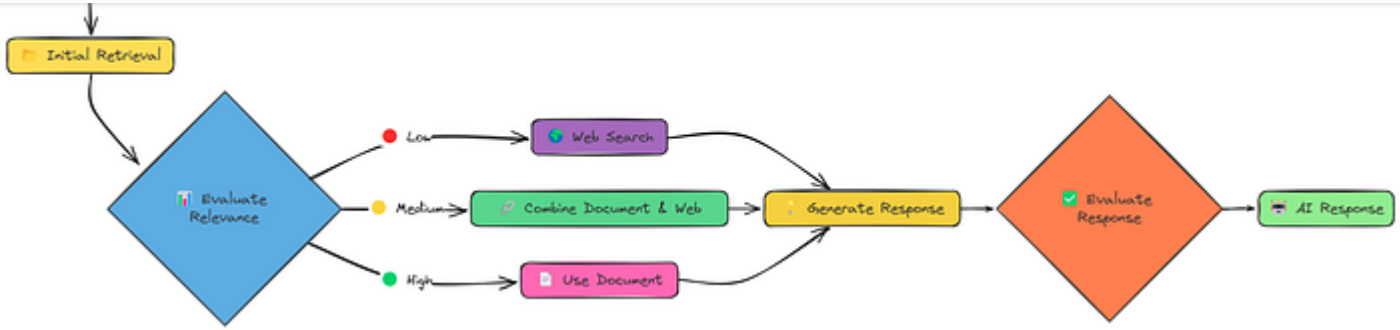
```
### Output
0.79
```

> We got a score of around 0.79.

Multi-modal RAG has the *potential* to be very powerful, especially for documents where images contain crucial information. However, it didn't beat the oter techniques that we have seen so far.

Crag

So far, our RAG systems have been relatively passive. They retrieve information and generate a response. But what if the retrieved information is bad? What if it's irrelevant, incomplete, or even contradictory? Corrective RAG (CRAG) tackles this problem head-on.



CRAG Workflow (Created by Fareed Khan)

CRAG adds a crucial step: *evaluation*. After the initial retrieval, it *checks* the relevance of the retrieved documents. And, crucially, it has different *strategies* depending on that evaluation:

- **High Relevance:** If the retrieved documents are good, proceed as usual.
- **Low Relevance:** If the retrieved documents are *bad*, fall back to a *web search*!
- **Medium Relevance:** If the documents are *okay*, combine information from both the document *and* the web.

This "corrective" mechanism makes CRAG much more robust than standard RAG. It's not just hoping for the best; it's actively checking and adapting.

Let's see how this works in practice. We'll use a function called `rag_with_compression` for this.

```
# Run CRAG
crag_result = rag_with_compression(pdf_path, query, compression_type="selective
```

This single function call does a *lot*:

1. **Initial Retrieval:** Retrieves documents as usual.
2. **Relevance Evaluation:** Scores each document's relevance to the query.
3. **Decision Making:** Decides whether to use the document, do a web search, or combine both.
4. **Response Generation:** Generates a response using the chosen knowledge source(s).

And, as always, the evaluation:

Copy

```
# Evaluate.
evaluation_prompt = f"User Query: {query}\nAI Response:\n{crag_result['response
evaluation_response = generate_response(evaluate_system_prompt, evaluation_pro
print(evaluation_response.choices[0].message.content)
```

> We're targeting a score of around 0.824.

CRAG's ability to *detect and correct* retrieval failures makes it significantly more reliable than standard RAG.

By dynamically switching to web search when necessary, it can handle a wider range of queries and avoid getting stuck with irrelevant or insufficient information.

This "self-correcting" ability is a major step towards more robust and trustworthy RAG systems.

Conclusion

The 18 tested RAG techniques represent diverse approaches to improving retrieval quality, from simple chunking strategies to advanced methods like Adaptive RAG.

While Simple RAG provides a baseline, more sophisticated approaches like Hierarchical Indices (0.84), Fusion (0.83), and CRAG (0.824) significantly

Adaptive RAG emerged as the top performer (0.86) by intelligently selecting retrieval strategies based on query type, demonstrating that context-aware, flexible systems deliver the best results across diverse information needs.

If you like reading this blog, you can follow me on Medium

#ai #artificial-intelligence #python #data-science #machine-learning